

CHOUDOJU AJITH KUMAR

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PROFESSIONAL SUMMARY

Final-year Electronics & Communication Engineering student specializing in embedded firmware development on ESP32/Arduino platforms. Skilled in designing non-blocking, state-machine-driven C/C++ firmware, integrating GPIO/I²C/SPI peripherals, and fusing multi-sensor data (IMU, RFID, ultrasonic) for real-time embedded systems. Complements hardware expertise with full-stack development (Next.js/React) and PCB design in Altium Designer to deliver end-to-end IoT and robotics solutions from schematic to cloud dashboard.

EDUCATION

Sreenidhi Institute of Science and Technology, Hyderabad

Sep 2023 – Jun 2027 (Expected)

B.Tech – Electronics and Communication Engineering | GPA: 7.17 / 10

EXPERIENCE

Cisco Networking Academy — Network Engineering Trainee, SNIST

2025

- Deployed and validated simulated enterprise network topologies (IP addressing, subnetting, static/dynamic routing, VLANs) in Cisco Packet Tracer, verifying connectivity through ping, traceroute, and CLI diagnostics; earned CCNA: Introduction to Networks certification.

General Secretary, Electronics Club, SNIST — CHARGE Revamped 2.0

Nov 2025 – Present

- Spearheaded planning and execution of a national-level technical symposium (Hackathon, Project Expo, Poster Presentation, Technical Games), directing cross-functional coordination between student committees, faculty, and volunteers across logistics, scheduling, and technical operations.

PROJECTS

Retrofit Smart Home Automation System — ESP32, C++, Firebase, Next.js/React, I²C, Altium Designer

- Architected a non-blocking, state-machine-driven ESP32 firmware (C/C++) to concurrently drive relay control, switch debouncing, PIR timing, and I²C OLED updates on a single core, eliminating blocking delays.
- Engineered bidirectional Firebase Realtime Database sync, ArduinoOTA updates, and a Next.js/React dashboard for remote appliance control; resolved cross-core race conditions between Firebase callbacks and the main loop via a dirty-flag synchronization pattern, and designed a custom retrofit PCB in Altium for integration with existing household wiring.

Multi-Robot Warehouse Navigation System — RFID, IMU, Ultrasonic Sensors, Wheel Encoders, BFS

- Engineered an autonomous warehouse robot combining RFID localization, IMU orientation tracking, ultrasonic obstacle detection, and encoder-based odometry for precision navigation.
- Designed a centralized BFS pathfinding algorithm for multi-robot coordination and collision avoidance, correcting cumulative IMU drift via periodic RFID checkpoints to hold positional error within one grid cell across full warehouse traversals.

Autonomous Secure Delivery Bot — Arduino, Fingerprint Sensor, Bluetooth — 3rd Place, Rampage v2.6 Hackathon

- Built an autonomous delivery robot with fingerprint-based biometric authentication for secure package access and Bluetooth-based manual override for fail-safe operation, delivering a fully functional hardware-software prototype in 24 hours as part of Team HackForge.

SKILLS

Languages: C, C++, Embedded C, Java, Python, SQL, JavaScript

Embedded & Hardware: ESP32, Arduino, GPIO, I²C, SPI, UART, Relay Drivers, Sensor Integration, PCB Design (Altium)

Frameworks & Tools: FreeRTOS, Firebase, Next.js, React, Node.js, Vercel, Cisco Packet Tracer

CERTIFICATIONS & ACTIVITIES

3rd Place – Rampage v2.6 Hackathon, KLH Bowrampally (2025) • Deloitte Cyber Job Simulation, Forage (Jul 2025) • Cisco CCNA: Introduction to Networks (2025) • Google for Developers – Build with AI Bootcamp, Hyderabad (Jul 2026)